

A QoS-Based routing algorithm over software defined networks

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ABSTRACT

Video streaming requires reliable transmission and a steady network to satisfy the video users and applications QoS needs. The network service must provide and meet a certain level of performance to accommodating these QoS, specially in Smart City environments, but current Internet best-effort service does not guarantee QoS. Software-defined networking (SDN) is an approach intended to separate the network control and data (forwarding) planes to get better management and optimization, which clearly enhance the planning and development of smart urban communities. In this paper, we propose a dynamic QoS algorithm over SDN to select the optimum path that meets video QoS in order to optimize the quality of experience (QoE). We differentiate the traffic based on video resolution QoE parameters to Standard Definition (SD) and High Definition (HD). The results demonstrate that the proposed method obtained better viewing quality for an smart community environment and increases the overall network throughput.

1. Introduction

Recently, there has been continuous growth in video consumption by Internet users, accompanied by the development in the technology of real-time video sessions, like IPTV and Internet video services. According to Seufert et al. (2015), the video traffic represented 57% of all Internet traffic in 2012, and probably this percentage has doubled these days due to the spread of social media and the diversity of video applications. Undoubtedly, video streaming requiring guaranteed performance which entails a particular requirement from the networks, those types of applications characterized as real-time (strict latency), rate-critical (specific data rate), and mission-critical (certain degree of reliability) applications (McCabe, 2010). Hence a good Quality of Service (QoS) is essential to stream video over the network and enhancing the Quality of Experience (QoE).

The Internet best-effort service created the lacking deployment of network QoS on Internet architecture since there are problems associated with current used short path routing. First, multiple metrics are hard to use in making routing decisions. Besides, it must consider only optimal cost paths, while cannot exploit the feasible routes. In contrast,

the QoS-based routing (QBR) concept appears to provide routing solutions that could achieve the quality needs. The basic idea is to determine network paths to various metrics to supply acceptable QoS for significant application flows, using some knowledge of resource availability in the network, also it must monitor and adapt with QoS parameters variations at network links (Costa and Duarte, 1999). In other words, QBR seeks to provide performance guarantees by using multiple metrics in terms of bandwidth, delay, packet loss, etc., as well, it could use best and non-best but acceptable paths (Crawley et al., 1998).

The various metrics used in path calculations could be handled as Single Mixed Metrics (SMM) or as multiple individual metrics (Wang and Crowcroft, 1996) (Costa et al., 2001). The first one computes and joins different QoS constraints in a single mathematical function, while the second applies many distinct metrics until it finds a feasible path that meets all quality restrictions. In this method, metrics composed by using three different rules, additive (e.g.: delay), multiplicative (e.g.: packet loss), and concave (e.g.: bandwidth). However, the SMM method cannot be sufficient for QoS routing, since it is uncertain that each QoS requirement is respected, as well as the difficulty of defining such function. Moreover, finding a path subject to multiple constraints is

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non-deterministic polynomial (NP)-complete problem, according to Wang and Crowcroft (1996). In this regard, a NP is a class of problems for which their solution can be found and verified in polynomial time. The algorithm used to solve them is called a non-deterministic algorithm. On the other hand, a problem is assumed to be NP-hard if every problem in NP can be reducible to it in polynomial time. A problem is NP-complete if it is both in NP and NP-hard, which means this problem is the hardest in NP. These types of problems are usually known as intractable. Conversely, the algorithm will be efficient if it can solve the problem in polynomial time. In contrast, if a given problem proves to be equivalent to an NP-complete structure, the solution will require exponential time and it is difficult to solve in polynomial time.

Furthermore, in traditional routing, source-based and hop-by-hop routing algorithms are proposed to achieve QoS routing. Source routing is based on the idea that the flow source is the one who calculates routes on-demand depending on the type of application, and it needs to have information of the entire network that necessary to generate the forwarding paths. On the other hand, hop-by-hop algorithms make routing decisions by using the information available at each router. It allows to reduce the setup delays and distribute the overhead, still, the routers might not be able to avoid the routing loops. However, the existing IP routing does not have the ability to verify the link QoS required to adapt with the requested requirements of QoS for the flow, since it is not aware of the available QoS over the path (Crawley et al., 1998).

The QoS model in the traditional networking faced unresolved issues such as per-hop decisioning, limited network global view, and difficulty of providing different QoS abilities for various application flows. On the other hand, compared to today's architectures, Software-Defined Networks (SDN) appeared to solve the problems of controlling and management of the networks, it decouple control plane and data plane and provide global network view, where controllers can obtain complete topology information and statistics. These features enhance QoS framework by:

- Applying Service Level Agreements (SLAs) and QoS management policies easily by the administrator, that can be changed dynamically at a higher abstraction level, without the need of setting them at each forwarding devices.
- The logically centralized controller, beside maintaining states information for the flow path could facilitate writing quality based routing algorithms and offer end-to-end QoS per flow.

Route decision in QoS routing is made corresponding to traffic characteristic, thus, the OpenFlow networks are a suitable choice, because their forwarding device tables are built on per flow bases. SDN can be very useful to state-dependent routing since the controller can re-route the video traffic through a different path in order to avoid congestion which enhance stream quality at the end receivers side.

This paper proposes a QoS-Based Routing model for video streaming over SDN. It includes the design of an algorithm that selects the optimal path among all paths available in the topology based on collecting the network status to detect the changes in video QoS metrics. Further, the model distinguishes the network traffic by using Differentiated Services Code Point (DSCP), where video flow is marked with the Assured Forwarding (AF) for multimedia streaming service classes, while best-effort flows take the default DSCP.

The rest of this paper is organized as follows. Next section includes the related work. The proposed video QoS-Based routing is described in Section III. The experimental setups and evaluation results are presented in Section IV. Section V depicted a comparison with other proposed QoS-based routing. Finally, conclusion and future work are shown in Section V.

2. Related work

Well-known techniques have been proposed to offer QoS, such as

integrated services (IntServ) (Braden et al., 1994), differentiated services (DiffServ) (Blake et al., 1998), and Multiprotocol Label Switching MPLS (Rosen et al., 1999). However, their implementations are limited due to the decentralized control of the current best-effort networks, resulting in static and complex architecture. Moreover, networking devices are more exposed to failures because of misconfigurations and missing of the automated settings (Egilmez et al., 2012).

On the other hand, several SDN routing QoS mechanisms have been proposed to support video streaming. H. E. Egilmez et al. designed the OpenQoS multimedia controller to provide a dynamic end-to-end QoS routing according to the network state (Egilmez et al., 2011). The controller enhances the video quality experienced by end-users by offering two paths. One is a QoS route specified for the multimedia streaming, and the second one is the shortest path for the other data. The route calculation metrics are the delay and the congestion and the available bandwidth. However, the used delay is fixed, which means that the dynamic route is built only on the gathered bandwidth statistics.

The architecture proposed in (Egilmez et al., 2012) provides an analytical optimization of the QoS routing problem, which is based on linear programming for scalable video coding (SVC). It aims to minimize the delay and offers no packet loss for QoS traffic. Likewise (Egilmez et al., 2013), and (Yan et al., 2015) proposed two optimization frameworks for SVC video stream and computed paths using the LARAC algorithm. The first study considers the SVC base layer as lossless flow with no packet loss while the enhancement layers could tolerate packet losses. In case of congestion, only the video base layer allows to reroute to other available non-shortest path routes. The second framework reroutes both base and enhancement layers to QoS-based routes, while the best effort traffic remains on its shortest path. These papers depended on the packet loss rate as a metric to determine the new routes. They assumed that if there is not sufficient available capacity, then, there will be packets loss. However, they do not consider the other possibilities of packet loss, such as security settings or attacks. They did not calculate the actual packet loss rate given by the obtained network state information. The study published in (Yan et al., 2015) used the delay variation as the constraint with packet loss rate to calculate QoS routes. Also, they ignored the influence of sufficient bandwidth as a routing metric. For video streaming, the delay variation could be improved by increasing the receiver buffers. Whereas, network bandwidth is often more critical for these applications, because if it is limited, this will lead to the queuing delay, and the packet loss rate will be higher, which will decrease the overall QoE of the end user.

Jinyao et al. (Yu et al., 2015) presented the HiQoS SDN framework to guarantee QoS. It is based on differentiated services and computes multiple paths between the source and destination. The framework distinguishes between different types of traffic by source IP address and categorizes user traffic into a video stream, interactive audio/video, and best-effort flow. Additionally, a modified Dijkstra algorithm was used to select the optimal path for the new flow according to the lowest bandwidth consumption. Using multiple paths enables HiQoS to be resistant to link failures through rerouting flow to another path. However, it only used the minimal bandwidth utilization of a queue as a metric, ignoring other video streaming quality metrics. Also, it classifies the flow with the source IP address, this might be not precise for video streaming, since different videos with different quality demands may be originated from the same device.

ARVS (Xu et al., 2015) suggested a two-level QoS routing approach for video streaming over SDN. They used the delay variation and cost to select the optimal paths for videos. Their work is like (Yan et al., 2015), except they utilized the shortest plus feasible QoS paths for all the traffic types. Both, the base layer video stream (level-1 QoS) and best-effort traffic, routed through the shortest path. ARVS will check this path against jitter constrain periodically, if it has jitter values over a threshold, the available bandwidth along the QoS path is examined, in case that the bandwidth is sufficient, the level-1 flow has priority to be re-routed to this path, and enhancement layer packets (level-2 QoS)

remained on the shortest path, while the level-2 re-routes to the QoS path and the base layer packets stay in the shortest path. However, packet loss is the most important constrained metric for a video stream at the network level and buffers at the application level can solve the jitter problem.

In the same direction (Sendra et al., 2017), offered a QoS-enabled management framework to support a queuing mechanism and route optimization algorithm to guarantee the transmission of the flow over SDN networks. It provides a suitable QoS for video streaming and multimedia applications, and also classifies the flows into different QoS levels. Their algorithm solves the constrained shortest path (CSP) problem based on the delay parameter. Besides, they compute the routing path cost with two additive metrics (delay and packet loss).

In 2017, authors of paper (Guo et al., 2021) presented a routing optimization system for SDN by applying reinforcement learning (RL) to enhance network QoS. The proposed routing protocol used this artificial intelligence (AI) method to select the optimal paths with the least cost in terms of delay, loss rate, and bandwidth. Additionally, Guo et al. (Civanlar et al., 2010) applied reinforcement learning (RL) in hybrid SDNs. It achieved link load balancing with the avoidance of routing loops by responding to dynamically changing traffic. Moreover, in (Liu et al., 2021), authors used a deep reinforcement learning (DRL) scheme to optimize routing in SDN of data-center networks. The method combines different network resources, such as bandwidth and cache memory, in order to minimize the delay. Then, it uses this information to improve the routing performance.

Rego et al. (2017) analyzed the effect of OSPF protocol on the network quality. Their study compared the protocol performance over SDN and traditional network. Another work, (Rego et al., 2019), took advantage of the SDN and AI to propose a dynamic routing metrics calculation for multimedia data. The presented proposal modified the OSPF protocol equation considering several metrics (bandwidth, delay, and loss). Besides, they implemented a messages exchange protocol between controller and switches to adapt the metrics according to the current topology state. Their results show that the delay and packet loss decreased, while bandwidth utilization increased.

Also a performance enhancement framework for IP video surveillance (IPVS) is presented in (Go et al., 2019). It takes advantage of SDN to adjust QoE bitrates and reroute traffic to maximize the resources available on the network. Employing rerouting ensures choosing the shortest and less loaded paths, so the video destination received the high-priority streams at superior quality. Framework results minimize packet loss, jitter, and latency, while it is also optimized the throughput.

Yamansavascilar et al. (Yamansavascilar et al., 2020) proposed a dynamic fault tolerance solution to improve the QoE of video streaming. They detect the congestion in the SDN link layer by using bidirectional forwarding detection (BFD) protocol, then, they employ data plane link failure tolerance mechanism to find alternative paths in order to offer good QoE.

Most of the previously mentioned studies focused on the CSP problem that tries to find the least cost path which satisfies only one constraint, while ignoring the multiple constrained shortest path (MCSP) problem. MCSP problem aims to find a path subject to multiple constraints. Video flow is one class of service that requires combinations of routing metrics to realize its QoS. Therefore, CSP does not estimate or guarantee that the path could fulfill all QoS constraints individually.

3. Video QoS-Based routing scheme

This section presents the main parts of the proposed video QoS-based routing approach.

3.1. Route constraint metrics selection

Video Streaming QoS requirements have to map into path metrics because the network should guarantee QoS relying on the definition of

the proper metrics. In general, finding the QBR path depends on many constraints such as additive (delay and jitter), multiplicative (packet loss), and concave (bandwidth) metrics. However, the computing complexity of choosing the best route can be very high, depending on the number of used metrics and their sequence. Wang and Crowcroft (1996) proved that finding a path subject to combinations of additive and multiplicative metrics is NP-complete if the number of metrics equals two or more metrics. Therefore, the choice of metrics and their count will depend on the importance of those metrics for the type of application (video streaming).

This work is concentrated on enhancing the QoE of two types of video streaming resolutions: High Definition (HD) and Standard Definition (SD) video resolutions. Therefore, we investigated the effect of different video QoS parameters that can influence the QoE and we focused on network-level related QoS parameters rather than video application-level parameters.

Video streaming is one of the most bandwidth demanding services, where its demands increases with the resolution of the video content. As a result, the bandwidth is more significant for video streaming, especially because higher video resolution means higher bitrate. Thus the network has to provide more capacity. Further, insufficient bandwidth will increase the delay and loss rate, leading to a decrease of end-user QoE. With high encoding bit rates, the stream is more exposed to packet loss, which has severe impact on the video streaming QoE. Users could experience frame freezing, complete loss of the video, or others problems depending on what video frames are lost. According to (ITU, 2008) and (Vega et al., 2018), the packet loss and bandwidth have a more effect on video streaming than jitter and delay. This happens because the jitter could be avoided by implementing de-jitter buffers, or by the playout buffers in the video application at the receiver side. Consequently, we applied packet loss and bandwidth as path metrics in our QoS-based routing method. Then, the impact of these QoS parameters on the quality of HD and SD video resolutions has been studied.

OpenFlow port counters are used to calculate the packet loss and bandwidth on each link. The controller obtains the switch port counters by sending the "OFPortStatsRequest" message. Then it receives the "OFPortStatsReply", which contains the received byte count and other statistical data.

3.2. QoS routing problem formulation

This study modeled the SDN network as a graph $G = (V, E)$, where V is a set of nodes, and $E = \{(i, j) : i, j \in V\}$ is a set of links. The path or route metrics are the packet loss rate ($PLR(r)$) and the available bandwidth ($\mathcal{A}(r)$). In order to obtain the $PLR(r)$ of the OpenFlow network, several factors should be considered, such as the OpenFlow port type if it is output source or destination ingress port, and the TX and RX buffers which are used to transmit and receive data. Depending on these factors, the PLR equation calculated based on TX buffers within output ports, and RX buffers with ingress ports, loss rate metric is associated with the specific link (i, j) between two network points, node i and node j , where $i = 1, \dots, n-1$ and $j = i+1, \dots, n$, and it is defined by:

$$PLR_{i,j} = \frac{(N_i TX_{output} - N_j RX_{input})}{N_i TX_{output}} \times 100 \quad (1)$$

where N is number of port, RX_{input} and TX_{output} are the set of buffers at ingress port and output port interface. Then the path loss probability between node s and node t approximated as:

$$PLR(r_{s,t}) = 1 - ((1 - PLR_{s,i}) \times (1 - PLR_{i,j}) \times (1 - PLR_{j,k}) \times \dots \times (1 - PLR_{l,t})) \approx$$

$$PLR(r_{s,t}) = PLR_{s,i} + PLR_{i,j} + PLR_{j,k} + \dots + PLR_{l,t} \quad (2)$$

Additionally, in order to compute available bandwidth metrics through an end-to-end path between source and destination, this work

needs to find the bandwidth utilization $\mathcal{B}U_{ij}$ and available bandwidth $\mathcal{A}B_{ij}$ for each link (i, j) between forwarding devices i and j on the topology. Also, this study assumed that the maximum capacity of every link in network topology C_{ij} is known, and the controller will estimate the link utilization after gathering network topology statistics. So, to find the bandwidth utilization, we suppose that the OpenFlow controller received `OFPortStatsReply` message at a time (t_1) , which contains the received (RB_{t_1}) bytes, and, after a period of interval (ΔT) , a second `OFPortStatsReply` received at the time (t_2) which has (RB_{t_2}) bytes, the $\mathcal{B}U_{ij}$ measured by bit/s calculated as follows:

$$\mathcal{B}U_{ij} = \frac{RB_{t_2} - RB_{t_1}}{\Delta T} \times 8 \quad (3)$$

Then, the available bandwidth for the link (i, j) is computed as:

$$\mathcal{A}B_{ij} = C_{ij} - \mathcal{B}U_{ij} \quad (4)$$

To sum up, the minimum unoccupied link bandwidth among all links (i, j) along that path determine the end-to-end path available bandwidth $\mathcal{A}(r)$. This link is known as the bottleneck, and its bandwidth is considered as the maximum possible bandwidth for the path, also recognized as the width path. $\mathcal{A}(r)$ for a path between node s and node t defined by the following equation:

$$\mathcal{A}(r_{s,t}) = \min [\mathcal{A}B_{s,i}, \mathcal{A}B_{i,j}, \mathcal{A}B_{j,k}, \dots, \mathcal{A}B_{l,t}] \quad (5)$$

Finally, if given two constraints, $\mathcal{B}\mathcal{T}$ (bandwidth threshold), and $\mathcal{P}\mathcal{L}\mathcal{T}$ (packet loss threshold), and there is a source, s , and a destination, $t \in V$, the routes between them are represented as a set of $\mathcal{R}(s, t)$. Then the QoS routing problem is to find a feasible route $r \in \mathcal{R}$, joined s and t , where:

$$\mathcal{A}(r_{s,t}) \geq \mathcal{B}\mathcal{T} \text{ and } \mathcal{P}\mathcal{L}\mathcal{R}(r_{s,t}) \leq \mathcal{P}\mathcal{L}\mathcal{T} \quad (6)$$

3.3. Processes description

By using terminologies defined in the previously, this research proposes a flow-based routing strategy for video streaming traffic subject to meet various QoS requirements. The primary goal of the routing strategy described below emphasizes to meet two routing constraints and efficiently determines two feasible paths for each video streaming flow on

the given current state of the network. Fig. 1 demonstrates the general paradigm of the QoS-based routing.

3.4. Proposed routing algorithm

Next we present the design of the routing algorithm in order to solve the QoS-based routing problem and achieve the optimization goal. There are two main ideas behind the design:

- The optimization criteria for the video streaming in this work is to minimize packet loss by choosing a path with lower end-to-end packet loss rate, but it can not exceed a specified threshold. Plus, maximize the capacity by choosing a path with high available bandwidth that meets video streaming bandwidth boundaries.
- The precedence between metrics is an important factor because it reflects the influence of network QoS parameters on overall QoE. Therefore, the proposed algorithm takes the bandwidth as secondary metric, and loss as primary one, since the latter is more impairing for video streaming, as it has been analyzed earlier, and there could be a route with high capacity but it can be losing the packets by different causes than congestion.

As discussed earlier, solving QoS problem depends on using multiple metrics to be NP-complete. This work proposes an heuristic algorithm based on the source routing algorithm presented by Wang et al. (Wang and Crowcroft, 1996). We considered it because of two features. Firstly, it computes forwarding paths on demand per flow basis, which is very appropriate for video streaming conditions. Secondly, in order to install the routing path, the entire network topology must be recognized, and this characteristic is offered by the SDN controller where it can access the full routing information of each link needed for the path computation. Our approach also includes in the algorithm the packet loss rather than delay, further finding two feasible paths based on k-short paths (KSP) Yen, Jin Y ((Yens and Yen, 1970)). rather than only Dijkstra ((Dijkstra, 1959)). In addition, it can find the widest path based on the bandwidth metric only; if the type of QoS policy permits it. In this regard, we have defined a policy rule to determine which traffic receives the QoS. It states that the network traffic must belong to one of three service groups: restricted QoS constraints called (group A), tolerable or soft QoS constraints (group B), and best-effort (group C). The flows in group A need enforcement of their QoS constraints, and those in group B

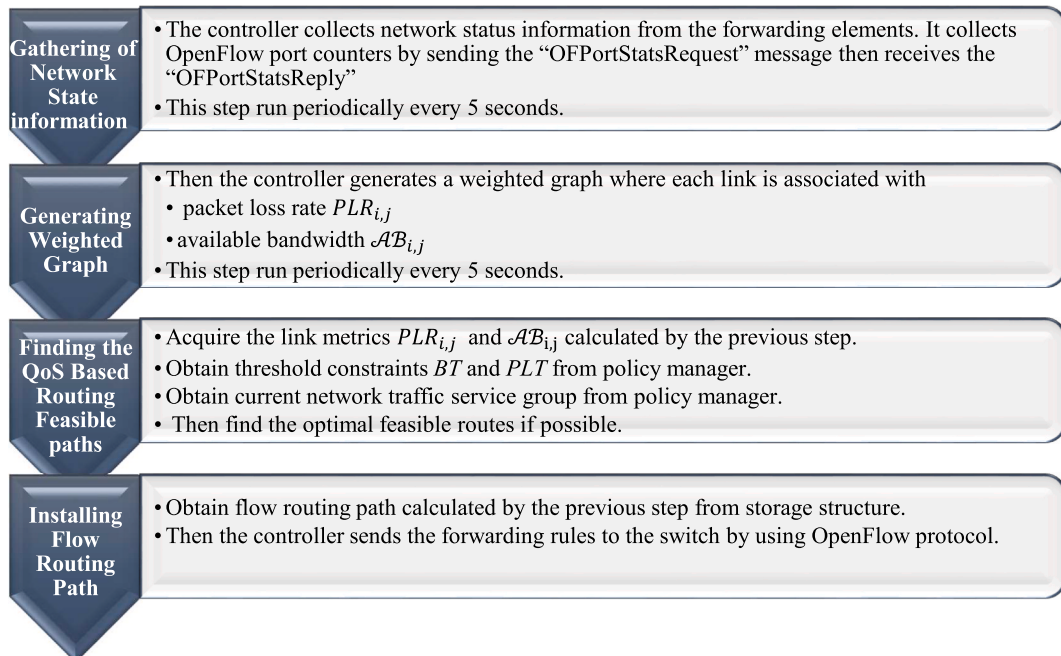


Fig. 1. Video QoS-Based routing steps.

are tolerant of acceptable performance guarantees, while group C does not claim any QoS guarantees. The proposed algorithm is called Two Lowest Loss -Widest Paths Algorithm (TwoLLWPs). It works as follows:

- There are two phases, elimination and search. First, all edges on original network topology graph that do not meet the bandwidth threshold are pruned. So, any paths in the resulting graph G satisfies $\mathcal{A}(r) \geq \mathcal{B}\mathcal{T}$. second, when there is more than one widest path meeting QoS bandwidth requirement, the algorithm begins to search two routing paths from the source node to the destination with the lowest packet loss rate $PLR(r) \leq \mathcal{P}\mathcal{L}\mathcal{T}$ if it possible based on Dijkstra algorithm and Yen's k-shortest path. The two paths must meet packet loss rate restriction. The path with lowest packet loss rate is called the lowest loss-widest-first path, while the second is called the lowest loss-widest-second alternative path.

- If at least one path is found, the flow admission status is changed to admitted, otherwise it is changed to "rejected".

- Furthermore, the algorithm does not have to find the minimum loss rate paths to all nodes, rather it finishes either when the destination node t is permanently identified or when the packet loss rate exceeds the threshold before reaching t .

- Yen's algorithm is used to determine the first k-shortest path by using Dijkstra to find the shortest path between two nodes. Then it starts to determine all other k-shortest paths. The T woLLWPs modifies Yen's algorithm by removing the first call of Dijkstra, because the first path is already computed.

- If it fails to find a feasible path meeting both, loss and bandwidth requirements, the algorithm returns the widest path based on bandwidth constraint by calling the Widest_Path_Dijkstra; if the policy rule associated with the flow allows this, because the biggest available bandwidth will be more desirable metric for video traffic rather than shortest path. The widest path is computed in one condition, if the application flows belong to group B, which has tolerable restrictions.

Fig. 2 shows the flowchart algorithm of process described before.

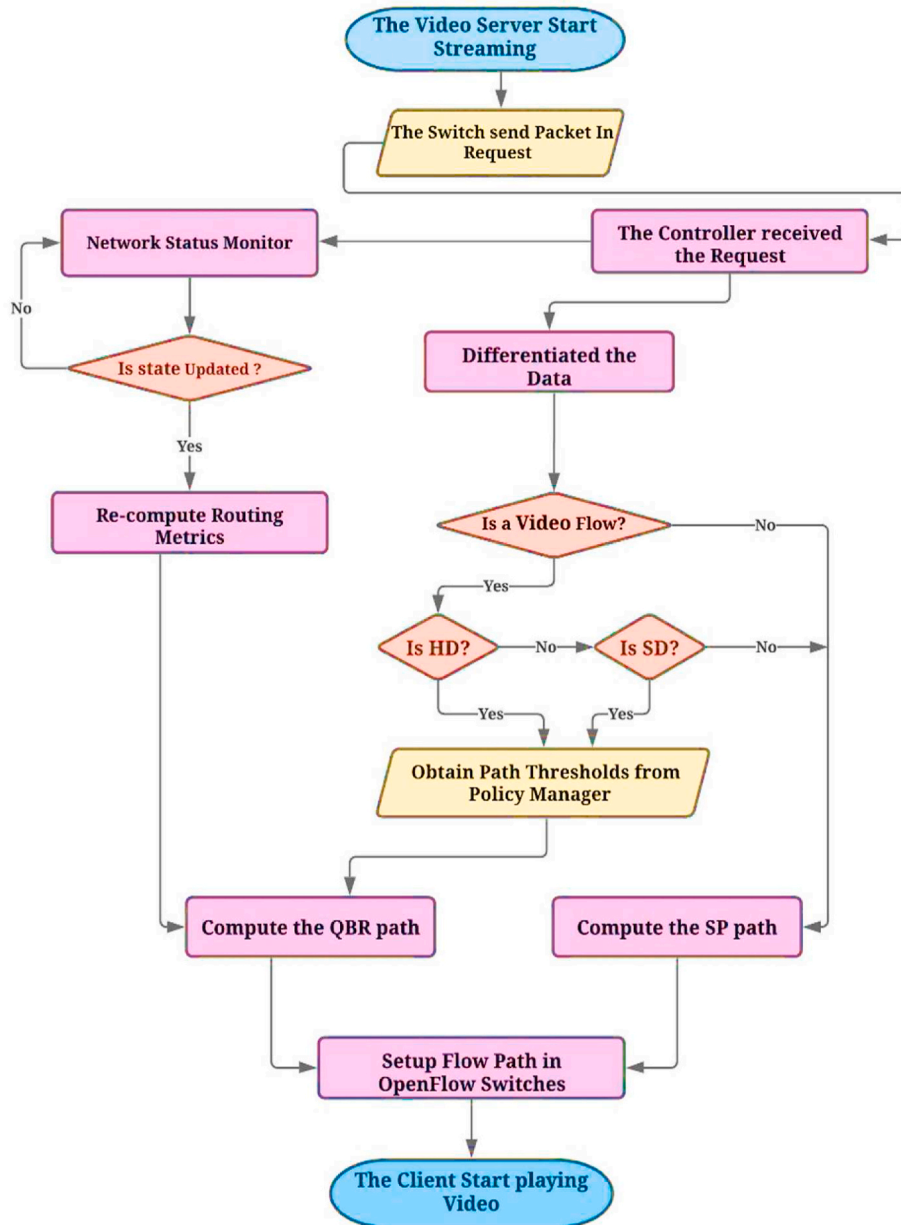


Fig. 2. Algorithm flow-chart.

Besides, all the procedures mentioned above are achieved by implementing these steps in [Algorithm 1](#).

Algorithm 1. Two Lowest Loss -Widest Paths (TwoLLWPs)

Input Weighed $G = (V, E)$, a graph with node and edge set, each edge has two weight values the $PLR_{i,j}$ and $\mathcal{B}_{i,j}$, values are non-negatives A source $s \in V$, a destination $t \in E$. Two constraints $\mathcal{B}\mathcal{T}$ and $\mathcal{P}\mathcal{L}\mathcal{T}$. Flow Group one of this values $FG : [A \text{ or } B]$.

Output HashMap with: paths set, path type, path counts and Flow Admission Status: FAS: [A for Admitted, R for Rejected].

Step 1: To create new G , $\forall i, j : PLR_{i,j} = \infty$ if $\mathcal{B}_{i,j} < \mathcal{B}\mathcal{T}$. // Prune phase

Step 2: $SPT = \{s\}$; $\forall i \neq s : PLR(p_i) = PLR(p_{s,i})$, $parents = \{\}$. // initialization

Step 3: Find $k \notin SPT$ so that $PLR_k = \min_{i \in SPT} PLR(p_i)$. If $PLR_k > \mathcal{P}\mathcal{L}\mathcal{T}$, $\{\}$. // No path could be found If $GF == B$, $\{r^* = \text{Widest_Path_Dijkstra}(G, s, t, \mathcal{B}\mathcal{T})\}$ If $r^* \neq \emptyset$, $\text{Build_HashMap}(r^*)$ Else $FAS = R$. $\}$ $FAS = R$. $\}$ End Algorithm. If $t \in SPT$, $\{\}$. // At least one path is found $FAS = A$. $r = \text{Build_Path}(parents, t)$ $KSP = \text{YenKSP}(newG, A, [0] = r, k = 2)$ // Call yen's algorithm to find second path If $KSPcount \geq 1$, $\text{Build_HashMap}(KSP)$, End Algorithm. $SPT := SPT \cup \{k\}$.

Step 4: $\forall i \notin SPT : PLR(p_i)^* := \min [PLR(p_i)^*, PLR_k + PLR_{k,i}]$, $parents[i] = k$.

Step 5: Go to Step 3.

The time complexity of this algorithm depends on its phases. The first phase prunes the graph by searching through all vertices and edges to eliminate those links that do not meet bandwidth constraints via replacing their packet loss rate by ∞ . This step is supposed to be executed once and requires $O(N^2)$ run time. Despite this, the floodlight controller already has a function to generate a weighted graph, so we modified it by inserting the “if” condition to keep only the links that fulfilled the bandwidth constraint. So, the graph is created from the beginning with only the desirable links. The second phase is to find two paths using Dijkstra and Yen algorithms. We implemented Dijkstra using a Fibonacci heap priority queue so it will run in $O(M + N \log N)$ time rather than $O(N^2)$, consequently Yen’s algorithm makes KN invocations of Dijkstra’s thus it takes $O(KN(M + N \log N))$ ([Mohanta and Poddar, 2012](#)), ([Bouillet et al., 2007](#)).

The proposed algorithm set $k = 2$, as a result, the complexity of using two algorithms gives equation (2):

$$\text{Complexity} = O(2N(M + N \log N)) \quad (7)$$

[Fig. 3](#) shows the messages exchanged between the OpenFlow switch, the SDN controller and the proposed QoS module. The message flow diagram includes the initial negotiation and the major messages between OpenFlow switch and the SDN controller. Whenever a server starts streaming, the switch sends a packet_In to the SDN controller. Then, it passes this request to the QoS module, which begins to set up an optimal path for video streaming.

4. Experimental results and analysis

This part presents the experimental assessment undertaken to evaluate the proposed solution. Different scenarios were designed to study its performance. The simulation testbed environment and the parameters are described. Additionally, the evaluation results are analyzed and discussed.

4.1. Testbed setup

[Table 1](#) describes the network parameters used for the experiments.

Table 1

General network parameters.

Topology Parameters	Values
Network links packet loss rates (%)	Ranging between: 0.001, 0.005, 0.05, 0.5, 1.0, and 2.0
Link bandwidth	Ranging between: 100 Mb/s, 10 Mb/s and 1 Mb/s
Number of links	22
Number of Nodes	1 Controller 9 Switch 6 Hosts (one of them is the Server)

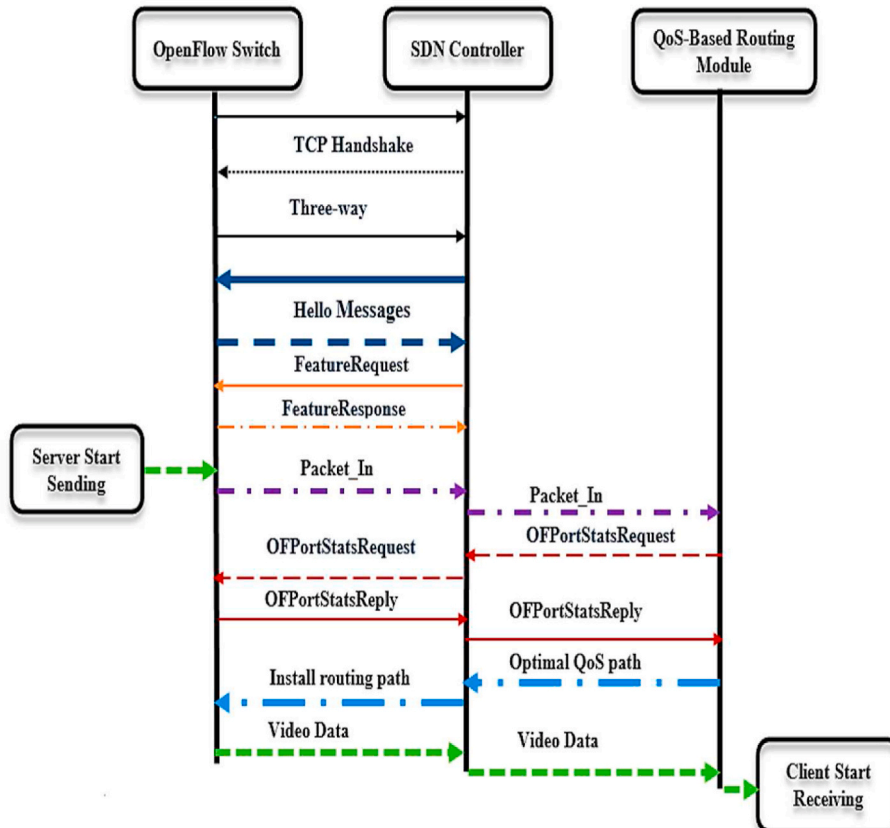


Fig. 3. Message flow diagram.

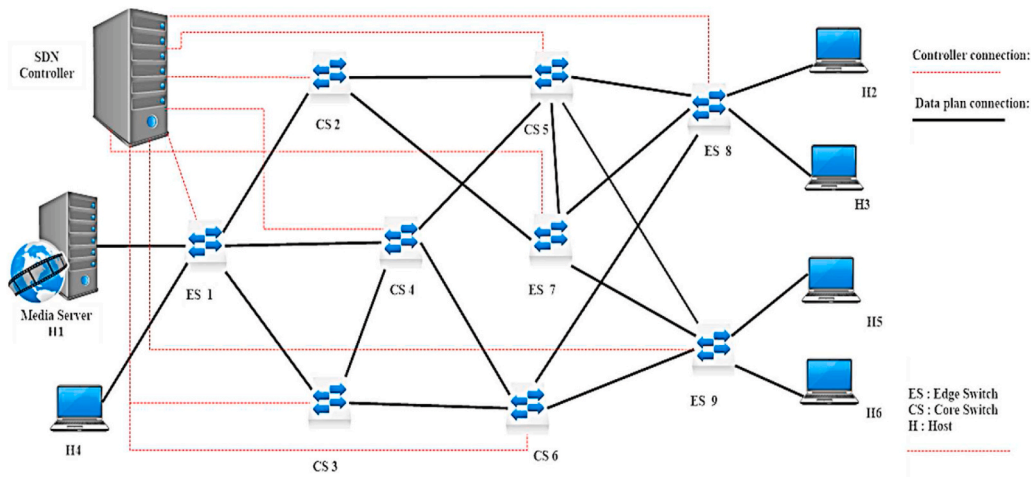


Fig. 4. General perspective of topology design.

Table 2
Used hardware and software.

Hardware or Software	Specifications
Ethernet Link Bandwidth	100 Mb/s
Computer OS Specifications	Microsoft Windows 10 Home, ×64-based PC System and Linux Ubuntu 18.04.5 LTS (Bionic Beaver) OS, ×64-based PC System
Programming Languages and Editors	• Java JDK 8. • Python v3.8. • Eclipse Editor. • PyCharm Editor.
Video System and Measurement Tools:	• GStreamer v1.18.4. • VLC Player. • Wireshark packet analyzer. • MSU Video Quality Measurement Tool 12.0 beta.

Simulation was performed using Mininet emulator. We implemented the network topology shown in Fig. 4. This figure is designed by SmartDraw software (SmartDraw, 2020).

The test was conducted using two laptops. One with Intel core™ i7-7500U, 2.70 GHz CPU, 2 Cores and 16 GB RAM. The other laptop with Intel core™ i7-4600U, 2.10 GHz, 2 Cores and 8 GB RAM. Also the setup included two virtual machines with two processors installed on Oracle VirtualBox Graphical User Interface Version 6.1.10. One of them contains the Floodlight controller master version and the second contains the Mininet Simulator version 2.3.0. Table 2 summarizes the used software and hardware specifications in both laptops.

4.2. Measurement of QoE metrics

Generally, video streaming QoE can be measured by subjective or objective methods, or both of them. Subjective metrics are conducted by asking subjects to rate the video they have been watching, the subjects rate the quality by excellent, good, fair, poor or bad. The Mean Opinion Score (MOS) includes well-known QoE scales, which is an average of scores across subjects, the MOS maps the ratings between excellent and bad to numerical scores between 5 and 1 (Rec, 2008) (Streijl et al., 2016).

The full reference (FR) model is used as an estimation method in video streaming QoE. It works by accessing both the original, called reference, and processed video, called distorted, to assess the quality. The properties of the two tested videos are compared frame-by-frame to check different features such as contrast features, and color processing. We use two types of FR measurement metrics for QoE estimation. The Structural Similarity Index Metric (SSIM) (Wang et al., 2004) and the Video Multimethod Assessment Fusion (VMAF) which has been recently

Table 3
SSIM and VMAF values mapped according to MOS scores.

MOS	SSIM	VMAF
Excellent = 5	>0.99	80–100
Good = 4	≥0.95 & < 0.99	60–79
Fair = 3	≥0.88 & < 0.95	40–59
Poor = 2	≥0.5 & < 0.88	20–39
Bad = 1	<0.5	<20

proposed by Netflix (Li et al., 2016). Table 3 shows an approximation objective QoE scales for SSIM and VMAF video quality metrics mapped to the MOS scores (Khan et al., 2008) (Zinner et al., 2010) (Li et al., 2018).

4.3. Proposed algorithm testing scenarios

Three test cases were designed to study the TwoLLWPs algorithm. It allows us to explain the importance of each QoS parameter and its impact. Their goals are as follows. The first test case compared the floodlight default mode (FDM) with video QoS mode (VQM), where in FDM all data traffic is routed by the original floodlight shortest path algorithm. In contrast, VQM transmits two types of data traffic, videos (combining two types, SD and HD) and best-effort. The video routed through the generated paths by TwoLLWPs algorithm, while the best-effort data remain at floodlight shortest paths. Second test case studies the various packet loss rate constraints influence on the video streaming QoE. Finally, third test case examines impact of different bandwidth constraints.

1) FDM vs VQM Test Case.

The SDN network Link bandwidth used in this test is 100 Mb/s, and the links of the longest paths have no loss, while other links take value > 0 and ≤ 2% loss metric. There is a background traffic that was generated using iperf testing tool between host 4 and host 6 (UDP traffic at 10 Mb/s). The test video samples used are the Sintel Trailer, obtained from Sintel website copyright under Blender Foundation (Sintel Trailer, 2021), and Caminandes videos under the Creative Commons Attribution 3.0 license from the caminandes.com (Caminandes, 2021). The policy rule categorizes the flow as belonging to the group (A). QoE and QoS parameters and their values are summarized in Tables 4 and 5.

Fig. 5 shows the variance between using FDM and the proposed video QoS-based routing. The fact of missing frames has a significant influence on FDM, which results in bad viewing experience compared to VQM because of using the shortest path rather than QoS routing. The video

Table 4

Video QoE parameters.

Video Name	Type SD or HD	Bitrate kbps	Size MB	Duration	Frame/ s
Sintel Trailer 480	SD	540	4.16	52 s	24
Sintel Trailer 1080	HD	2116	13.9	52 s	24
Caminandes_llamigos_1080p	HD	1042	191	2:30 m	24

Table 5

Network QoS parameters.

Video Type	FDM	VQM
		Packet Loss Rates Constrain Metrics
Class 28 (SD)	–	0.5%
Class 30 (HD)	–	0.005%
		Bandwidth Constrain Metric
		3 Mb/s
		10 Mb/s

artifacts problems seen in FDM include annoyance and blocking in frame pixels. Also, sometimes complete image losses are shown. The frame in VQM is relative to the original frame with few color distortions.

Fig. 6 and Fig. 7 show the impact of the TwoLLWPs and FDM on different received video sequences resolutions. X-axis represents tested frames and the Y-axis represents the structural similarity index metric (SSIM). As observed, the video QoE is affected by QoS network parameters. In this experiment, the video without the QoS-based routing scheme reflected a dropping in the perceived video quality. For SD type, the obtained SSIM values for Sintel (480) video indicate higher quality viewing behavior. In addition, HD performance results also increased when it is compared to shortest path routing.

In addition, Fig. 8 displays the mean VMAF scores for Sintel Trailer samples. The results exhibit that the proposed approach offers high scores, which indicates good viewing quality for SD and fair for HD.

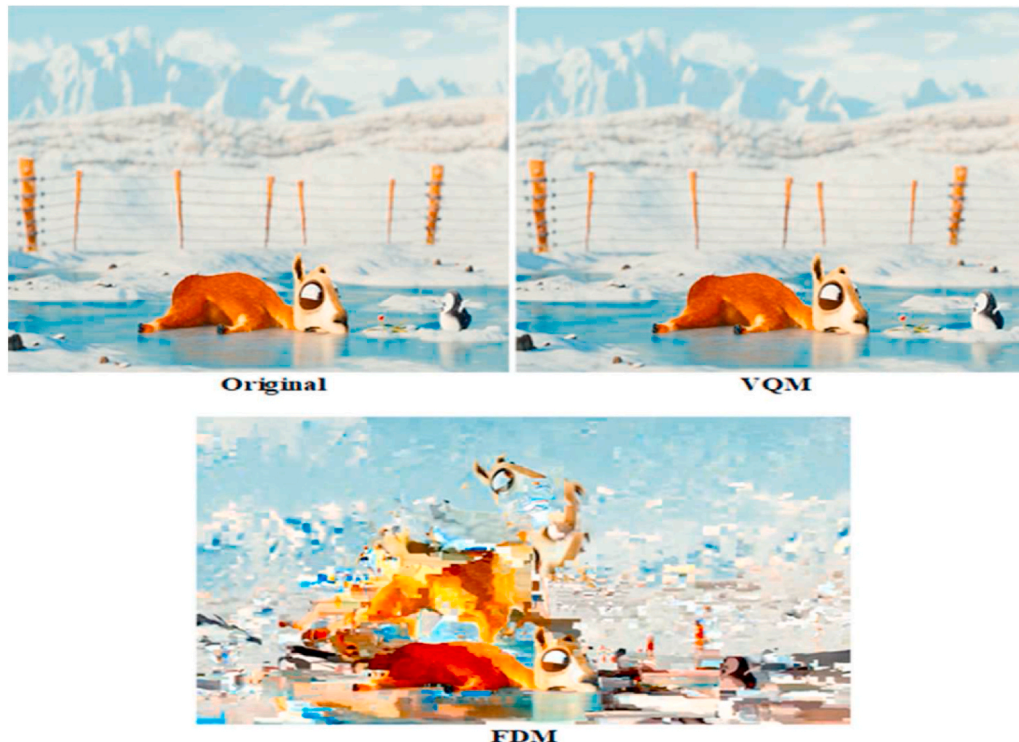
While FDM's highest average value does not exceed 30 for SD, for HD value is less than 10, which suggests a poor, and bad QoE perception.

Also, we investigated the network throughput achieved in the case of using FDM and VQM. This time we used the Caminandes Llamigos video (with a resolution of 1920×1080) and Wireshark packet analyzer. Fig. 9 presents the video traffic throughput for FDM, and Fig. 10 shows the network throughput received for the proposed model. As noted in the first figure, the throughput decreases significantly due to the packet loss increase, where the maximum value reaches only 2500 bytes per second. Conversely, the second figure shows that VQM achieves a higher throughput, where the maximum bytes per second range between 400,000 and 500,000; the reason is that the TwoLLWPs reduces the packet loss probability and gives information of the available bandwidth, which improves the network throughput.

Many remarks can be summarized from the above results. For example, the number of video frames received by VQM is higher than the number of frames received by FDM because the lossy links have been avoided with the proposed QoS-based routing algorithm. Moreover, the HD video resolution produces lower objective SSIM and VMAF metrics ratings in comparison to SD resolution, because HD frames contain more information than SD, therefore in case of packet loss, the HD incurs more degradation in the video viewing experience.

2) Influence of Packet Loss Rate as Constraint.

This test seeks to study the effect of different packet loss values as constraints on the proposed routing algorithm. The parameters deployed in the experimental set-up are the following ones. The video samples used for this test were Sintel Trailer SD and HD, and the policy rule categorizes the flow as belonging to the group (B). Link capacity is set to 10 Mb/s. The links of the longest paths are divided into two parts: (i) part with no loss, and (ii) part with 0.3% loss, while other take value > 0 and $<$ loss metric. The background traffic is set to be about 3 Mb/s. The loss rates constraint metrics used for this test are: 0.001, 0.005, 0.05, 0.5, 1.0, and 2.0, while the bandwidth constraints metrics are fixed to 5 Mb/s for SD and 10 Mb/s for HD. For the FDM, the metrics settings in

**Fig. 5.** Original Video vs FDM and VQM.

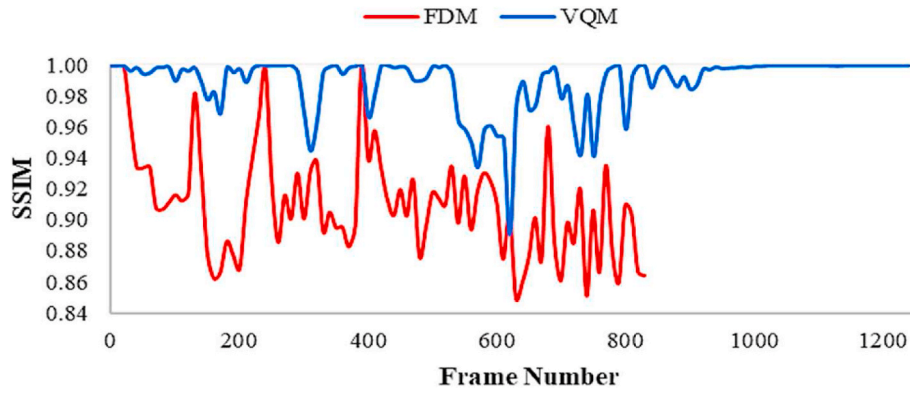


Fig. 6. FDM vs VQM impact on SD Videos.

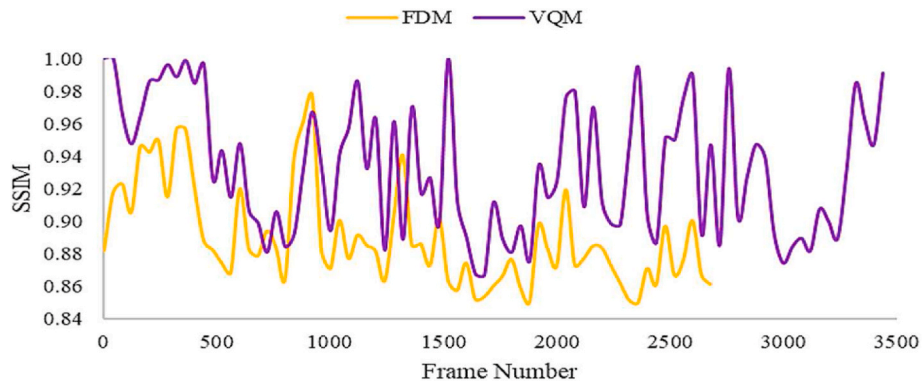


Fig. 7. FDM vs VQM impact on HD Videos.

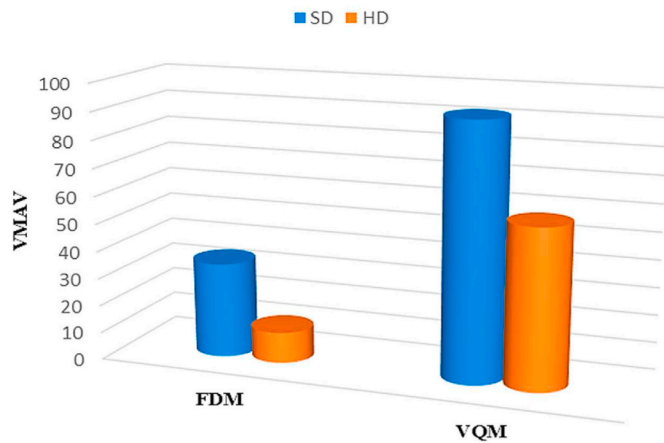


Fig. 8. Average values of VMAF Objective Quality Metric.

floodlight have no effect. Finally, the video quality was measured by SSIM.

Fig. 11 presents a sample of received HD video by the TwoLLWPs with different loss rates metrics. FDM is compared with the original video. The results suggest that the QoE of the coded video increased when using the proposed algorithm. As noticed, even with metric up to 2% PLR, the perceived impairments are less than the default shortest path.

Figs. 12 and 13, and Fig. 14 show the impact of different packet loss rate values on SD and HD videos. Fig. 12 shows that when the PLR = 0.05%, SSIM decreases. A similar result happens when PLR = 1%. When PLR = 0.5%, the video has higher SSIM values. The degradation of

quality when we use small loss rate values is due to the probability of rejection by the algorithm because it takes more time to find feasible paths. If it fails to find one, it takes the width path since the policy rule used in this test configures the flow to belong to group B, which has soft constraints, and the widest is considered acceptable for this group. However, this reduced SSIM. In contrast, higher loss rates initially result in higher scores as the probability of finding paths meeting the two constraints increases. However, the video quality significantly improves after a while when the controller reads the new network states.

In a similar manner to SD video, Fig. 13 presents the SSIM values measured during HD video delivery, when the PLR = 0.005%, the SSIM decreases at the beginning but improves later, and when PLR = 0.5%, SSIM scores are higher at the start of video reception but, it dropped afterward. It is explained using the same reasons aforementioned in the analysis of SD results. Moreover, when the packet loss rate is high, it indicates a degraded received QoS, e.g., when PLR = 1%, PLR = 2%. Finally, Fig. 14 shows that when the values of packet loss rate metrics are so close to each other, there is not much difference in the observed quality. Accordingly, as can be seen from the figures, choosing a loss metric that is not too small will enhance the performance of the routing algorithm; this must, however, be in balance with selecting the appropriate threshold.

3) Influence of Bandwidth as Constraint

In order to evaluate bandwidth metric impact on the performance of the routing algorithm, different bandwidths have been applied (1, 5, 10 and 20 Mb/s). The link capacity set to 50 Mb/s, and two hosts generate about 20% UDP best-effort-traffic. The longest paths have no loss and the rest take values > 0 and $\leq 2\%$, and the loss input metrics are 0.5% and 0.05% for SD and HD respectively.

Fig. 15 presents the obtained results. It shows that badwidth

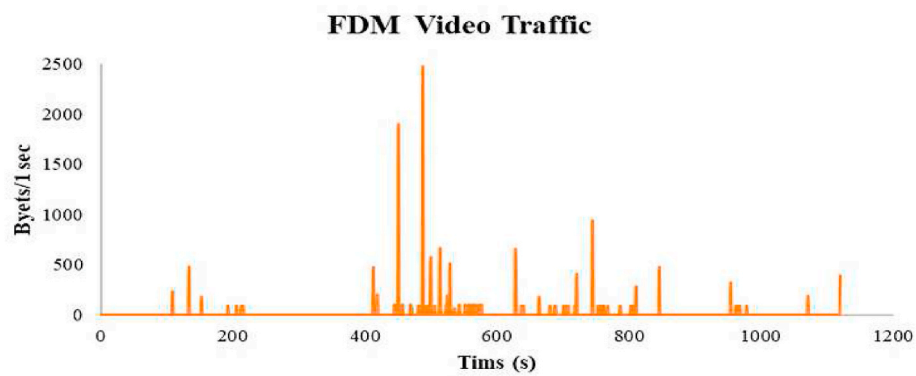


Fig. 9. FDM network throughput.

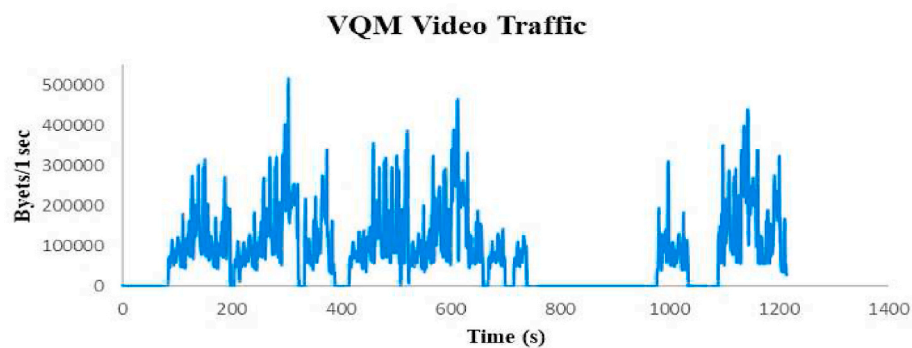


Fig. 10. VQM network throughput.



Fig. 11. Impact of setting the TwoLLWPs by packet loss rates (0.005% or 2%) against FDM.

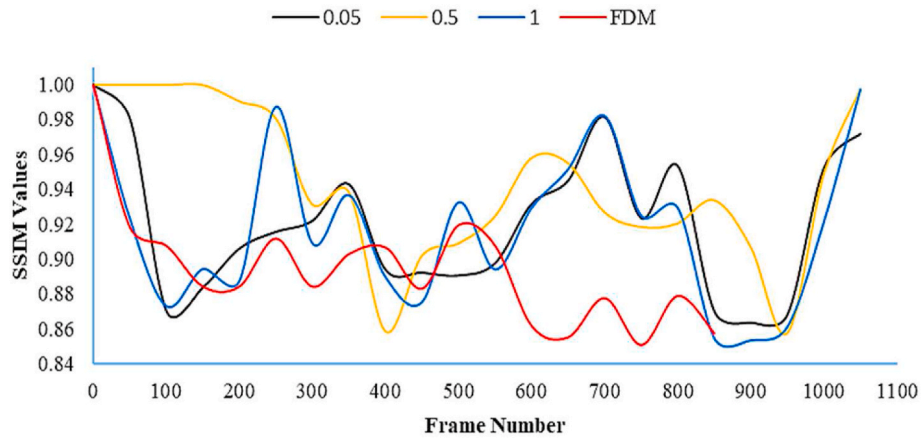


Fig. 12. Impact of many PLR values on QoE of SD video.

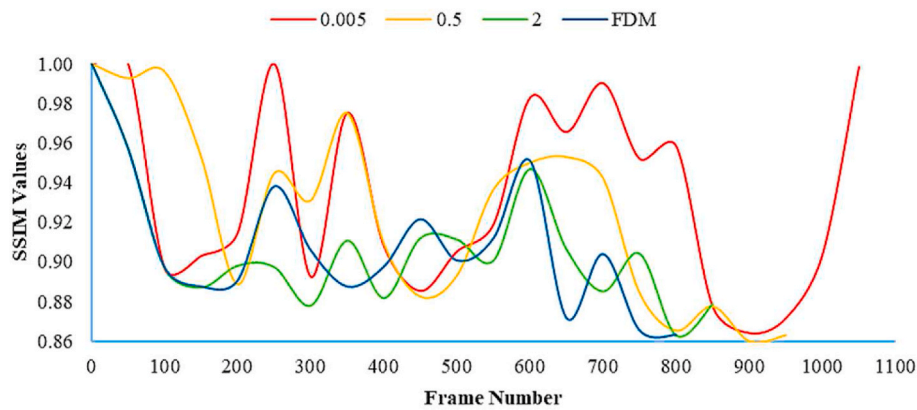


Fig. 13. Impact of a Number of PLR on QoE of HD video.

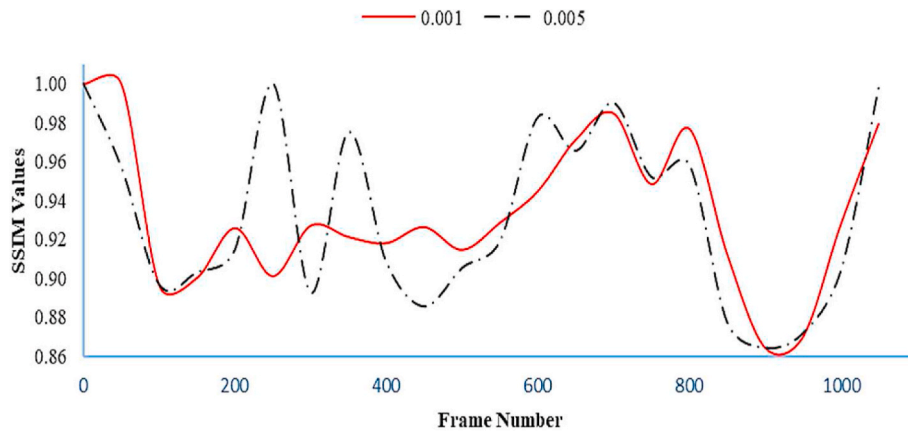


Fig. 14. Impact of Several PLR (0.001 and 0.005) on QoE of HD video.

constraints strongly affects the videos. As we can see, in Fig. 15, the differences in the results are irregular; e.g., in SD, the average SSIM of 10 Mb/s obtains better results than that of 20 Mb/s, despite that the 20 Mb/s means the chosen path is supposed to have a rate higher than 20 Mb/s. We can explain the behavior by the fact that the selected route ultimately depends on the loss metric; for example, the admitted path rate for the threshold = 10 might be 40 Mb/s, and for the threshold = 20, it might be 22 Mb/s. As well, the same observations are given in the case of HD videos. Moreover, due to the best effort that competes for resources, the quality of videos may have dropped since QoS-based routing

lacks a mechanism to reserve resources. Finally, small bandwidth metrics increase the options for routing paths, while high bandwidth metrics reduce these options and increase rejection rates.

5. Discussion

Table 6 illustrates a comparison between the proposed approach and other QoS-based routing proposals. The majority of the studies are focused on achieving only one constraint of video streaming (Civanlar et al., 2010) (Egilmez et al., 2012) (Yu et al., 2015), (Xu et al., 2015),

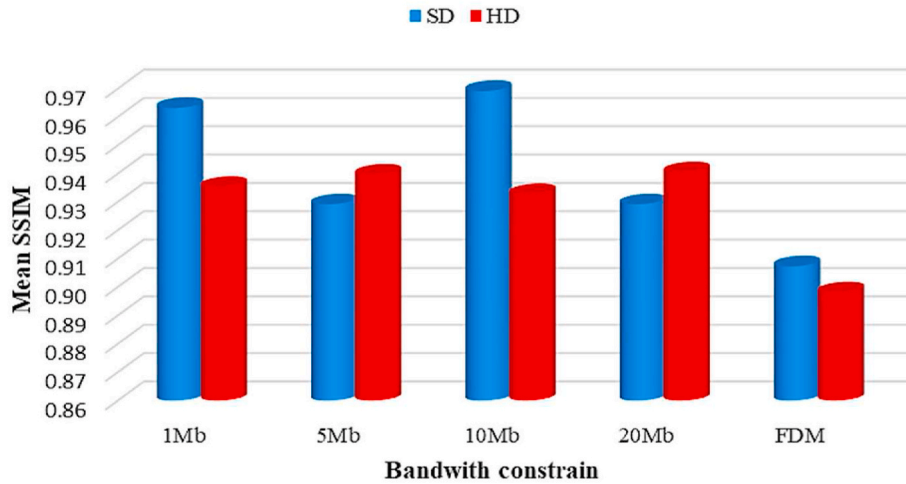


Fig. 15. Impact of several bandwidths on QoE of SD and HD Video.

Table 6

Comparison of related work with the proposed solution on QoS-Based routing.

#	QoS-Based Routing Algorithm	Routing Metrics	The applied Constraints	Video Parameters	QoS/QoE Measurement metrics
Civanlar (Egilmez et al., 2012)	The Solution not Developed	Available Bandwidth	Delay	SVC B/E layer	PSNR
Egilmez (Yan et al., 2015)	LARAC	Two mixed metrics: Packet loss and jitter	Jitter	SVC B/E layer	PSNR
Jinyao (Yu et al., 2015)	Modified Dijkstra	Two mixed metrics: Delay and Bandwidth	Minimal Bandwidth Utilization	N/D	Server Response Time and Throughput
Yu (Xu et al., 2015)	LARAC	Two mixed metrics: Jitter and Packet loss	Jitter	N/D	PSNR
Xu (Sendra et al., 2017)	Random Discretization Algorithm	Two mixed metrics: Packet loss and Delay	Delay	N/D	Packet Loss, Jitter and Throughput
Rego (Go et al., 2019)	Dijkstra	Three mixed metrics: Packet loss, Bandwidth and Delay	AI Dynamical Cost Equation to Specify the Constraint between: Loss, Bandwidth and Delay	N/D	Bandwidth Utilization, Packet Loss, Jitter and Delay
Our proposal	TwoLLWPs	Multiple Individual metrics: Available Bandwidth and Packet loss rate	Multiple individual Metrics: Bandwidth and Packet loss rate	SD, HD Ready, HD	SSIM, VMAF and Throughput

and), while some others are focused on optimizing the performance without addressing specific QoS requirements (Yan et al., 2015)() and (Rego et al., 2019)(). Similarly, many of them use the cost of route selection based on a single mixed metric, and, as we stated previously, when it depends on the SMM, it does not guarantee each QoS parameter individually. Alternatively, our proposal, TwoLLWPs, meets multiple individual QoS metrics.

Moreover, the effective way to evaluate the quality of the video is by considering the user perspective and assessment, which is known as QoE. It is achieved by implementing subjective tests or by objective QoE metrics. However, three studies (Yan et al., 2015; Xu et al., 2015; Rego et al., 2019) evaluated the video quality based on network-level performance measurements. Additionally, (Civanlar et al., 2010; Egilmez et al., 2012), and (Yu et al., 2015) measured the video quality via Signal-to-Noise Ratio (PSNR), but it does not reflect the human perception. For this reason, it cannot be considered a reliable method for assessing QoE (Huynh-Thu and Ghanbari, 2012). Instead, we use SSIM and VMAF, which can analyze blurring, luminance, contrast, global noise, as well as blocking and color distortions to detect artifacts that can be perceived by human eyes.

6. Conclusion and future work

In this paper, we proposed a flow-based routing strategy for video

streaming. The primary goal of the routing strategy emphasized meeting two routing constraints, packet loss, and bandwidth. It efficiently determines feasible paths for each video flow according to the current state of the whole network, like in a smart city. We took advantage of SDN to implement our state-dependent routing scheme by benefiting from the controller global view of the network. It can be used for video surveillance and video delivery in smart cities and smart urban communities. Simulation results have shown that the solution mechanism offers higher video QoE. Moreover, the results suggest that HD videos are impacted more than SD video when there is more packet loss, because it obtains more frame artifacts and color distortions.

This study focuses on investigating packet loss and bandwidth metrics associated with HD and SD. Although the proposed routing algorithm depends on thresholds defined by the administrator, therefore it is possible to implement any other QoS requirements related to video streaming, such as the 4 K videos resolution, if the policy defines their loss and bandwidth metrics.

More work is needed to support video QoS-based routing, such as QoE policy enforcement and dynamic resource sharing to provide a method for requesting and reserving network resources for the selected routing paths. Further, we will deploy and investigate the proposed algorithm across inter-domains.

Declaration of competing interest

Authors declare that they have no conflict of interest.

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